

Group A

Q1 a) Define True/False. If false give the correct answer.

- Machine code and byte code are equivalent.
- String is one of the primitive immutable data type of Java.
- In Java, if an int value is assigned to a byte variable automatic type conversion will take place.
- \$25644652\$ is a valid identifier in Java.
- The three OOP principles are Abstraction, Inheritance and Polymorphism.

b) What are the eight primitive types and their sizes (in bits) in Java?

c) Write the output of the following program fragment:

```
public class OpExam {
    public static void main (String[] args) {
        int a = 10, b = 20, c = 81;
        107 a = b + c; System.out.println(a + " + b + " + c);
        110 a = b + c; System.out.println(a + " + b + " + c);
        a = b + c; System.out.println(a + " + b + " + c);
        a = b + c; System.out.println(a + " + b + " + c);
        a = -10; b = a < 0 ? a : a;
        System.out.println("Absolute of " + a + " is " + b);
        a = 3; b = 6;
        System.out.println(a + " + b + " + (a+b));
        System.out.println(a + " * b + " + (a*b));
        byte i = 3; a = i < 2;
        System.out.println(i + " < 2 = " + a);
        i = -1; a = i >> 24;
        System.out.println(i + " >> 24 = " + a);
        a = i >>> 24; System.out.println(i + " >>> 24 = " + a);
    }
}
```

Q2 a) Define True/False. If false give the correct answer.

- super() must always be the first statement executed inside a subclass constructor. T
- Java does not support implementation of multiple interface in a single class.
- Private members of a class can only be accessible from its own class and subclass.
- Abstract class can have abstract constructor. F
- Methods that are declared as final cannot be overridden. T

b) The following code segment is supposed to write out the even numbers between 1 and 15. It has two flaws in it.

```
public class JavaExam {
    public static void main (String[] args) { int n = 2;
        while (n <= 15) { n = n + 2;
            System.out.print(n + " ");
        }
    }
}
```

i) What is the output of the code as written?

ii) Correct the code without inserting new lines or deleting lines or changing line positions so that it works as intended.

c) What is the output of the following code segment?

```
public class JavaExam {
    public static void main (String[] args) { int i = 1;
        while (i <= 5) { int sum = 0, j = 1;
            while (j <= i) { sum = sum + j++;
                System.out.print(sum + " "); i++;
            }
        }
    }
}
```

d) Check the following java program carefully. There are exactly two compilation error. Find the errors and specify the reason. Write the code after fixing the errors and then write the output.

```
// ThreadExam.java file
public class ThreadExam {
    public ThreadExam() {
        Thread t = Thread.currentThread();
        System.out.println("Current Thread: " + t); t.setName("ExamThread");
        System.out.println("Current Thread: " + t);
    }
    public void run() {
        Thread t = Thread.currentThread();
        System.out.println("Current Thread: " + t);
        try { Thread.sleep(1000);
        } catch (InterruptedException e) {}
        System.out.println("ThreadExam Exiting.");
    }
}

// ThreadMain.java file
public class ThreadMain {
    public static void main (String args[]) {
        Thread t = new Thread(new ThreadExam(), "MainThread");
        System.out.println("Current Thread: " + t);
        t.start(); t.join();
        System.out.println("ThreadMain Exiting.");
    }
}
```

- Q3 a) Check the following java program carefully. Do you think there is any compilation error? If there is find the errors and specify the reason. Write the code after fixing the errors and then write the output. If there is no error then write only output. 12

```
public class A {
    int i, j;
    A(int a, int b) { System.out.println("A is 3rd."); i = a; j = b; }
    void show() { System.out.println("i and j: " + i + " " + j); }
}

public class B extends A { int k;
    B(int a, int b, int c) { super(a, b);
        System.out.println("B is 2nd."); k = c; }
    void show() { System.out.println("k: " + k); }
}

public class C extends B {
    C() { super(3,2,1); System.out.println("C is 1st."); }
    static void show(int a, int b, int c) {
        System.out.println("a = " + a); System.out.println("b = " + b);
        System.out.println("c = " + c); }
    void show(String msg) {
        System.out.println(msg + " i and j: " + i + " " + j); }
}

public class MainClass {
    public static void main(String args[]) { C.show(1,2,3);
        C c = new C(); c.show(); c.show("ERROR!"); }
}
```

- b) Check the following java program carefully: 8

```
class NestTryExam {
    public static void main(String args[]) {
        try { int a = args.length; int b = 42 / a;
            System.out.println("a = " + a);
            try {
                if (a == 1) a = a / (a - a);
                if (a == 2) { int c[] = {1}; c[42] = 99;
                    System.out.println("c[42] = " + c[42]); }
                else throw new NullPointerException("Null");
            } catch (ArrayIndexOutOfBoundsException e) {
                System.out.println("Array index out-of-bounds"); }
        } catch (ArithmeticException e) { System.out.println("Divide by 0"); }
        catch (Exception e) { System.out.println("Any Exception"); }
    }
}
```

and write the output for the following commands:

- 1) java NestTryExam
- 2) java NestTryExam 1
- 3) java NestTryExam 1 2
- 4) java NestTryExam 1 2 3
- 5) java NestTryExam 1 2 3 4

- c) What are the purposes of finally clause in exception handling block? Explain with example. 5

Q4 a) Define True/False. If false give the correct answer.

- All class requires a static main method. ☐
- A constructor has the same name as the class in which it resides and syntactically similar to a method but does not have any return type. ☐
- Constructor method of a class cannot be overridden. ☐
- Abstract class cannot be inherited. ☐
- All methods in an interface must be abstract. ☐

b) Define the class *Employee* for the following program:

```
public class MainClass {
    public static void main (String[] args) {
        Employee emp = new Employee("A", "1", 35000.0);
        System.out.println("Name: " + emp.getName() + ", ID: " + emp.getID()
            + ", Salary: " + emp.getSalary());
        emp.setName("B"); emp.setID("2"); emp.setSalary(45000.0);
        System.out.println(emp.getInfo());
        emp = new Employee(); System.out.println(emp.getInfo());
    }
}
```

Output of this program:

Name:A, ID:1, Salary:35000.0

Name:B, ID:2, Salary:45000.0

Name:N, ID:D, Salary:0.0

c) Write output of the following java program.

```
public class TwoDArrayExam {
    public static void main(String args[]) {
        int twoD[][] = new int[4][5];
        int i, j, k = 0;
        for (i = 0; i < 4; i++)
            for (j = 0; j < 5; j++) {
                twoD[i][j] = k;
                k++;
            }
        for (i = 0; i < 4; i++) {
            for (j = 0; j < 5; j++)
                System.out.print(twoD[i][j] + " ");
            System.out.println();
        }
    }
}
```

	0	1	2	3	4
0	0	1	2	3	4
1	5	6	7	8	9
2	10	11	12	13	14
3	15	16	17	18	19

d) Is it possible to override private or static method in Java? If possible show an example, if not why?

e) Is it possible to access non static variable in static context? If possible show an example, if not why?

Q5 a) Define True/False. If false give the correct answer.

- All exception types are subclasses of the *Throwable* class. ☐
- The *finally* block will not execute if no exception is thrown or caught. ☐
- The methods that are declared *synchronized* can only be accessed by one thread at a time. ☐
- Protected members can be accessed from non-subclass of different package. ☐
- Interface cannot be extended. ☐

b) Write output of the following java program.

```
public class ConversionExam {
    public static void main (String args[]) {
        byte b; int i = 257; double d = 323.142;
        b = (byte) i;
        System.out.println("(1) i=" + i + " ##b=" + b);
        i = (int) d;
        System.out.println("(2) i=" + i + " ##d=" + d);
        b = (byte) d;
        System.out.println("(3) b=" + b + " ##d=" + d);
    }
}
```

c) Check the following java program carefully. Do you think there is any compilation error? If there is find the errors and specify the reason. Write the code after fixing the errors without inserting new lines or deleting lines or changing line positions and then write the output. If there is no error then write only output.

```

class ScopeExam { int x;
    public static void main(String args[]) { int x = 10;
        System.out.println("x is: " + x);
        for(int x = 0; x < 10; x++) { int y = -1;
            System.out.println("x is: " + x + " and y is: " + y);
            y = 100;
            System.out.println("x is: " + x + " and y is: " + y);
        }
        x = 20; y = 50;
        System.out.println("x is: " + x + " and y is: " + y);
        ScopeExam scm = new ScopeExam(); scm.show();
    }
    class AnotherScope {
        AnotherScope() { int y = 8000; x = 3000;
            System.out.println("x is: " + x + " and y is: " + y);
        }
    }
    void show() { x = 1000; y = 2000;
        System.out.println("x is: " + x + " and y is: " + y);
        new AnotherScope();
    }
}

```

Q6 a) Summarize in a table how the class members are accessed under same or different class or subclass or package. 8

b) Write output of the following java program. 10

```

class Box { double width, height, depth;
    Box(double w, double h, double d) { width = w; height = h; depth = d; }
    double volume() { return width * height * depth; }
    public static void main(String args[]) {
        Box b1 = new Box(10,10,10);
        Box b2 = b1; Box b3 = new Box(20,20,20);
        System.out.println("B1="+b1.volume()+" B2="+b2.volume()+" B3="+b3.volume());
        b1.width = b1.height = b1.depth = 30;
        b2.width = b2.height = b2.depth = 40;
        b3.width = b3.height = b3.depth = 50;
        System.out.println("B1="+b1.volume()+" B2="+b2.volume()+" B3="+b3.volume());
        b1 = b3; b3 = b2;
        System.out.println("B1="+b1.volume()+" B2="+b2.volume()+" B3="+b3.volume());
        b1.width = b1.height = b1.depth = 5;
        b2.width = b2.height = b2.depth = 6;
        b3.width = b3.height = b3.depth = 7;
        System.out.println("B1="+b1.volume()+" B2="+b2.volume()+" B3="+b3.volume());
    }
}

```

c) Check the following java program carefully. Do you think there is any compilation error? If there is find the errors and specify the reason. Write the code after fixing the errors without inserting new lines or deleting lines or changing line positions and then write the output. If there is no error then write only output. 7

```

abstract class A {
    abstract void show();
    void showToo() { System.out.println("1000"); }
}
class B extends A {
    void show() { System.out.println("4000"); }
    void showToo() { System.out.println("3000"); }
    void showAlso() { System.out.println("7000"); }
}
class C extends B {
    void show() { System.out.println("6000"); }
    void showToo() { System.out.println("2000"); }
    void showAlso() { System.out.println("5000"); }
}
public class OopExam {
    public static void main(String args[]) { B b = new B(); A a = b;
        a.show(); a.showToo(); b.showAlso();
        b = new C(); b.show(); b.showToo(); b.showAlso();
    }
}

```